

Initiating Search

November 6, 2025, 4:48 PM

• Search:

CAS Lexicon Search:

Transfer hydrogenation - Preferred Concept

Search Tasks

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Returned Reference Results from CAS Lexicon (3,070)	 References	View Results
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Filtered By:		
Yield:	90-100%, 80-89%	
Document Type:	Journal	
Language:	English	
Publication Year:	2020 to 2025	

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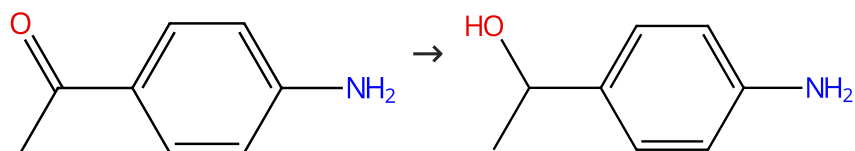
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (102)

 Suppliers (64)

31-513-CAS-21446217

Steps: 1 Yield: 100%

Strong solvent effects on catalytic transfer hydrogenation of ketones with [Ir(cod)(NHC)(PR₃)] catalysts in 2-propanol-water mixtures

1.1 Catalysts: Potassium *tert*-butoxide, 2409182-34-7
Solvents: Isopropanol, Water; 1 h, 80 °C

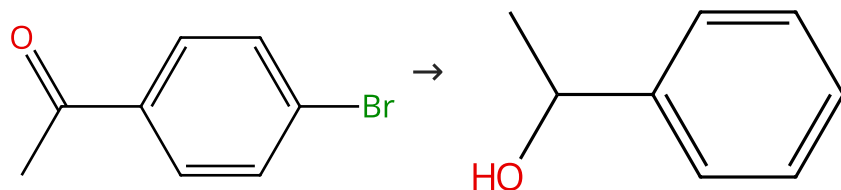
By: Orosz, Krisztina; et al

Catalysts (2020), 10(1), 17pp..

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (84)

 Suppliers (71)

31-113-CAS-20975200

Steps: 1 Yield: 100%

Hetero-Bimetallic Complexes Based on an Anthryridine Ligand Preparation and Catalytic Activity

1.1 Reagents: Sodium hydroxide
Catalysts: 2400891-28-1
Solvents: Isopropanol; 20 h, 80 °C

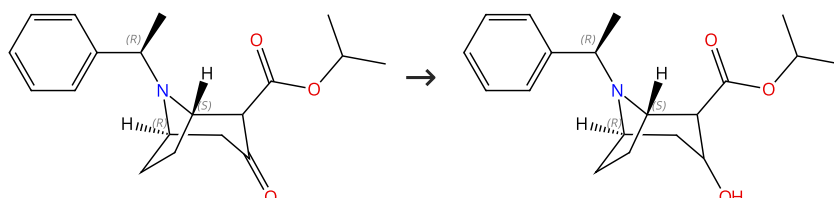
By: Lin, Shih-Chieh Aaron; et al

Organometallics (2020), 39(1), 123-131.

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

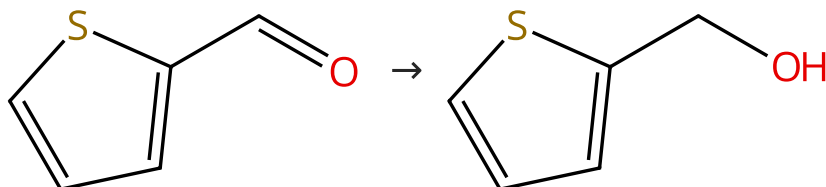
Absolute stereochemistry shown

 Supplier (1)

31-614-CAS-31175557	Steps: 1 Yield: 100%	Development of a Scalable Route with Efficient Stereoisomer Control to YZJ-1139, an Orexin Receptor Antagonist
1.1 Reagents: Sodium borohydride Solvents: Methanol; 2 h, -10 °C		By: Zeng, Zhenya; et al
Experimental Protocols		Organic Process Research & Development (2022), 26(2), 447-457.

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



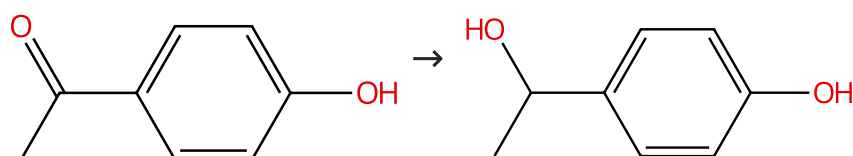
Suppliers (98)

Suppliers (81)

31-614-CAS-35498685	Steps: 1 Yield: 100%	Catalytic Base-Free Transfer Hydrogenation of Biomass Derived Furanic Aldehydes with Bioalcohols and PNP Pincer Complexes
1.1 Reagents: Ethanol Catalysts: (OC-6-13)-Carbonyl[2-(diphenylphosphino-κP)-N-[2-(diphenylphosphino-κP)ethyl]ethanamine-κM][tetrahydroborato(1-)-κH]ruthenium; 30 min, 130 °C		By: Padilla, Rosa; et al
Experimental Protocols		ChemCatChem (2023), 15(2), e202200819.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



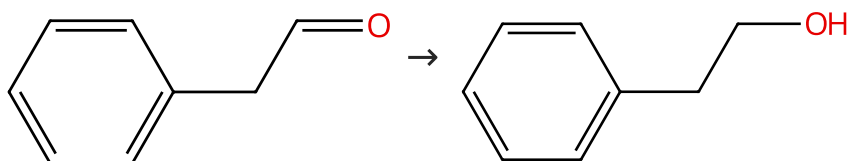
Suppliers (122)

Suppliers (73)

31-513-CAS-22960076	Steps: 1 Yield: 100%	Synthesis and characterization of Pd(II) and Ru(II) complexes of tetradentate N,N,N,N-(diphosphinomethyl)amine ligands: catalytic properties in transfer hydrogenation and heck coupling reactions
1.1 Reagents: Isopropanol, Potassium carbonate Catalysts: 2561423-50-3; 12 h, 82 °C		By: Urus, Serhan; et al
Experimental Protocols		Heterocycles (2020), 100(7), 1019-1034.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



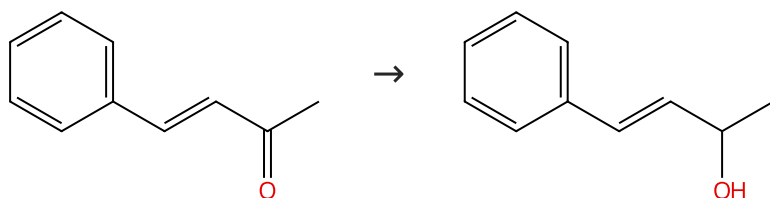
Suppliers (69)

Suppliers (110)

31-614-CAS-39254884	Steps: 1 Yield: 100%	Construction of Cu-Ru bimetallic catalyst for the selective catalytic transfer hydrogenation of carbonyl (C=O) in biomass-derived compounds
1.1 Reagents: Isopropanol Catalysts: Zirconium dioxide, Sulfur trioxide, Silica, Copper alloy, nonbase, Cu,Ru; 12 h, 180 °C		By: Cai, Bo; et al
Experimental Protocols		Renewable Energy (2024), 222, 119833.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



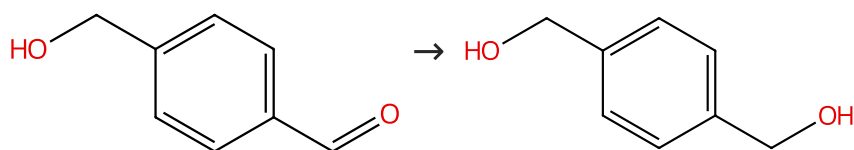
Suppliers (70)

Suppliers (26)

31-614-CAS-45409336	Steps: 1 Yield: 100%	Manganese Atom-Induced Lattice Distortion in Fe ₃ O ₄ to Boost Selective Transfer Hydrogenation of Unsaturated Oxygenates
1.1 Catalysts: Iron manganese oxide (Fe _{2.1} Mn _{0.9} O ₄) Solvents: Isopropanol; 8 h, 200 °C		By: Zhang, Jiaxing; et al
		ACS Catalysis (2025), 15(9), 7505-7515.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



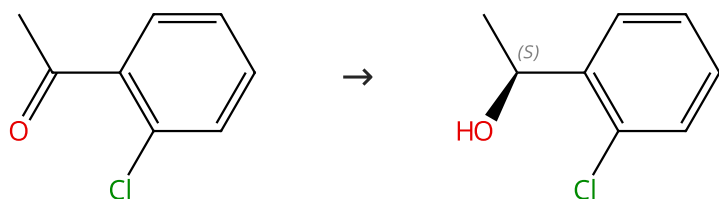
Suppliers (71)

Suppliers (89)

31-513-CAS-22344501	Steps: 1 Yield: 100%	Fluorescent and Biocompatible Ruthenium-Coordinated Oligo (p-phenylenevinylene) Nanocatalysts for Transfer Hydrogenation in the Mitochondria of Living Cells
1.1 Reagents: Sodium formate Catalysts: Ruthenium(1+), [[[4-[[[2-(amino-κM)methyl]amino-κM]sulfonyl]phenyl]methyl][3-[[6-[4-methoxy-2,5-bis(2-phenylethenyl)phenoxy]hexyl]amino]-3-oxopropyl]diphenylphosphoniumato]chloro[(1,2,3,4,5,6-η)-1-methyl-4-(1-methylethyl)benzene]-, bromide (1:1) Solvents: DMSO- <i>d</i> ₆ , Water- <i>d</i> ₂ ; 12 h, 37 °C		By: Dai, Nan; et al
Experimental Protocols		Chemistry - A European Journal (2020), 26(20), 4489-4495.

Scheme 9 (10 Reactions)

Steps: 1 Yield: 100%



Suppliers (85)

Absolute stereochemistry shown,
Rotation (-)

Suppliers (68)

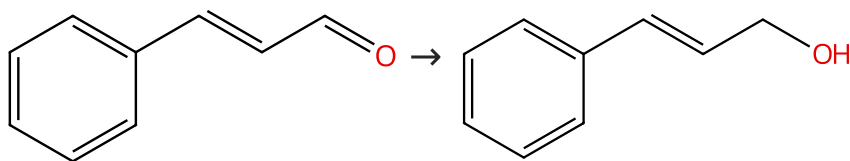
31-614-CAS-36774281 Steps: 1 Yield: 100% 1.1 Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Dichlorotris (triphenylphosphine)ruthenium Solvents: Isopropanol; 2 h, 85 °C 1.2 Reagents: Triethylamine Catalysts: (αR)-α-(1-Methylethyl)benzo[<i>h</i>]quinoline-2-methanamine Solvents: Isopropanol; 6 h, 85 °C; 85 °C $\rightarrow$ 23 °C 1.3 Reagents: Potassium <i>tert</i>-butoxide Solvents: Isopropanol, <i>tert</i>-Butanol; 2 h, 23 °C Experimental Protocols	Pincer Ru with a single stereo genic identity for highly efficient asymmetric transfer hydrogenation of ketones By: Xu, Xingjun; et al Science China: Chemistry (2023), 66(5), 1443-1449.
31-614-CAS-36774285 Steps: 1 Yield: 100% 1.1 Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Dichlorotris (triphenylphosphine)ruthenium Solvents: Isopropanol; 2 h, 85 °C 1.2 Reagents: Triethylamine Catalysts: (αR)-α-Methyl-4-phenylbenzo[<i>h</i>]quinoline-2-methanamine Solvents: Isopropanol; 6 h, 85 °C; 85 °C $\rightarrow$ 23 °C 1.3 Reagents: Potassium <i>tert</i>-butoxide Solvents: Isopropanol, <i>tert</i>-Butanol; 45 min, 23 °C Experimental Protocols	Pincer Ru with a single stereo genic identity for highly efficient asymmetric transfer hydrogenation of ketones By: Xu, Xingjun; et al Science China: Chemistry (2023), 66(5), 1443-1449.
31-614-CAS-36774293 Steps: 1 Yield: 100% 1.1 Catalysts: Dichlorotris(triphenylphosphine)ruthenium, 1,1'-Bis [bis(4-methoxyphenyl)phosphino]ferrocene Solvents: Isopropanol; 2 h, 85 °C 1.2 Reagents: Triethylamine Catalysts: (αR)-α-(1,1-Dimethylethyl)-4-phenylbenzo[<i>h</i>]quinoline-2-methanamine Solvents: Isopropanol; 6 h, 85 °C; 85 °C $\rightarrow$ 23 °C 1.3 Reagents: Potassium <i>tert</i>-butoxide Solvents: Isopropanol, <i>tert</i>-Butanol; 20 min, 23 °C Experimental Protocols	Pincer Ru with a single stereo genic identity for highly efficient asymmetric transfer hydrogenation of ketones By: Xu, Xingjun; et al Science China: Chemistry (2023), 66(5), 1443-1449.
31-614-CAS-36774288 Steps: 1 Yield: 100% 1.1 Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Dichlorotris (triphenylphosphine)ruthenium Solvents: Isopropanol; 2 h, 85 °C 1.2 Reagents: Triethylamine Catalysts: (αR)-α-Cyclohexyl-4-phenylbenzo[<i>h</i>]quinoline-2-methanamine Solvents: Isopropanol; 6 h, 85 °C; 85 °C $\rightarrow$ 23 °C 1.3 Reagents: Potassium <i>tert</i>-butoxide Solvents: Isopropanol, <i>tert</i>-Butanol; 1 h, 23 °C Experimental Protocols	Pincer Ru with a single stereo genic identity for highly efficient asymmetric transfer hydrogenation of ketones By: Xu, Xingjun; et al Science China: Chemistry (2023), 66(5), 1443-1449.

31-614-CAS-36774286 Steps: 1 Yield: 100% 1.1 Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Dichlorotris (triphenylphosphine)ruthenium Solvents: Isopropanol; 2 h, 85 °C 1.2 Reagents: Triethylamine Catalysts: (αR)-α-(1-Methylethyl)-6-phenyl-2-pyridineme thanamine Solvents: Isopropanol; 6 h, 85 °C; 85 °C $\rightarrow$ 23 °C 1.3 Reagents: Potassium <i>tert</i>-butoxide Solvents: Isopropanol, <i>tert</i>-Butanol; 2 h, 23 °C Experimental Protocols	Pincer Ru with a single stereo genic identity for highly efficient asymmetric transfer hydrogenation of ketones By: Xu, Xingjun; et al Science China: Chemistry (2023), 66(5), 1443-1449.
31-614-CAS-36774283 Steps: 1 Yield: 100% 1.1 Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Dichlorotris (triphenylphosphine)ruthenium Solvents: Isopropanol; 2 h, 85 °C 1.2 Reagents: Triethylamine Catalysts: (αR)-α-(1,1-Dimethylethyl)-4-phenylbenzo[<i>h</i>]quinoline-2-methanamine Solvents: Isopropanol; 6 h, 85 °C; 85 °C $\rightarrow$ 23 °C 1.3 Reagents: Potassium <i>tert</i>-butoxide Solvents: Isopropanol, <i>tert</i>-Butanol; 45 min, 23 °C Experimental Protocols	Pincer Ru with a single stereo genic identity for highly efficient asymmetric transfer hydrogenation of ketones By: Xu, Xingjun; et al Science China: Chemistry (2023), 66(5), 1443-1449.
31-614-CAS-36774280 Steps: 1 Yield: 100% 1.1 Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Dichlorotris (triphenylphosphine)ruthenium Solvents: Isopropanol; 2 h, 85 °C 1.2 Reagents: Triethylamine Catalysts: (αR)-α-(1-Methylethyl)-4-phenylbenzo[<i>h</i>]quinoline-2-methanamine Solvents: Isopropanol; 6 h, 85 °C; 85 °C $\rightarrow$ 23 °C 1.3 Reagents: Potassium <i>tert</i>-butoxide Solvents: Isopropanol, <i>tert</i>-Butanol; 1 h, 23 °C Experimental Protocols	Pincer Ru with a single stereo genic identity for highly efficient asymmetric transfer hydrogenation of ketones By: Xu, Xingjun; et al Science China: Chemistry (2023), 66(5), 1443-1449.
31-614-CAS-36774287 Steps: 1 Yield: 100% 1.1 Catalysts: Dichlorotris(triphenylphosphine)ruthenium, 1,1'-Bis [bis(4-fluorophenyl)phosphino]ferrocene Solvents: Isopropanol; 2 h, 85 °C 1.2 Reagents: Triethylamine Catalysts: (αR)-α-(1,1-Dimethylethyl)-4-phenylbenzo[<i>h</i>]quinoline-2-methanamine Solvents: Isopropanol; 6 h, 85 °C; 85 °C $\rightarrow$ 23 °C 1.3 Reagents: Potassium <i>tert</i>-butoxide Solvents: Isopropanol, <i>tert</i>-Butanol; 30 min, 23 °C Experimental Protocols	Pincer Ru with a single stereo genic identity for highly efficient asymmetric transfer hydrogenation of ketones By: Xu, Xingjun; et al Science China: Chemistry (2023), 66(5), 1443-1449.

31-614-CAS-36774278 Steps: 1 Yield: 100% 1.1 Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Dichlorotris(triphenylphosphine)ruthenium Solvents: Isopropanol; 2 h, 85 °C 1.2 Reagents: Triethylamine Catalysts: (αR)-4-Methoxy-α-(1-methylethyl)benzo[<i>h</i>]quinoline-2-methanamine Solvents: Isopropanol; 6 h, 85 °C; 85 °C $\rightarrow$ 23 °C 1.3 Reagents: Potassium <i>tert</i>-butoxide Solvents: Isopropanol, <i>tert</i>-Butanol; 1 h, 23 °C Experimental Protocols	Pincer Ru with a single stereogenic identity for highly efficient asymmetric transfer hydrogenation of ketones By: Xu, Xingjun; et al Science China: Chemistry (2023), 66(5), 1443-1449.
31-614-CAS-36774284 Steps: 1 Yield: 100% 1.1 Catalysts: Dichlorotris(triphenylphosphine)ruthenium, 1,1'-Bis[bis(4-chlorophenyl)phosphino]ferrocene Solvents: Isopropanol; 2 h, 85 °C 1.2 Reagents: Triethylamine Catalysts: (αR)-α-(1,1-Dimethylethyl)-4-phenylbenzo[<i>h</i>]quinoline-2-methanamine Solvents: Isopropanol; 6 h, 85 °C; 85 °C $\rightarrow$ 23 °C 1.3 Reagents: Potassium <i>tert</i>-butoxide Solvents: Isopropanol, <i>tert</i>-Butanol; 30 min, 23 °C Experimental Protocols	Pincer Ru with a single stereogenic identity for highly efficient asymmetric transfer hydrogenation of ketones By: Xu, Xingjun; et al Science China: Chemistry (2023), 66(5), 1443-1449.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (52)

Suppliers (86)

31-614-CAS-35127546 Steps: 1 Yield: 100% 1.1 Reagents: Isopropanol Catalysts: Zirconium dioxide; 2 h, 120 °C	Fabrication of Zirconium-Doped Activated Carbon by Chemical Activation for Catalytic Transfer Hydrogenation of 5-Hydroxymethylfurfural into 2,5-Dihydroxymethylfuran By: Wang, Yue; et al Energy & Fuels (2022), 36(22), 13796-13807.
Experimental Protocols	

Scheme 11 (2 Reactions)

Steps: 1 Yield: 100%



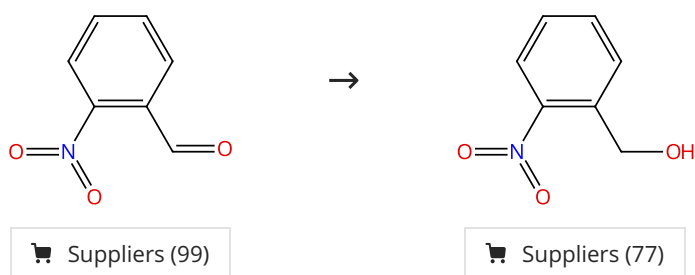
Suppliers (85)

Suppliers (71)

31-513-CAS-21446215 Steps: 1 Yield: 100% 1.1 Catalysts: Potassium <i>tert</i>-butoxide, 2409182-34-7 Solvents: Isopropanol, Water; 1 h, 80 °C Experimental Protocols	Strong solvent effects on catalytic transfer hydrogenation of ketones with [Ir(cod)(NHC)(PR₃)] catalysts in 2-propanol-water mixtures By: Orosz, Krisztina; et al Catalysts (2020), 10(1), 17pp..
31-513-CAS-22964821 Steps: 1 Yield: 100% 1.1 Reagents: Isopropanol, Potassium carbonate Catalysts: 2561423-54-7; 12 h, 82 °C Experimental Protocols	Synthesis and characterization of Pd(II) and Ru(II) complexes of tetradentate N,N,N,N-(diphosphinomethyl)amine ligands: catalytic properties in transfer hydrogenation and heck coupling reactions By: Urus, Serhan; et al Heterocycles (2020), 100(7), 1019-1034.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-37500197 Steps: 1 Yield: 100% 1.1 Reagents: Isopropanol Catalysts: Potassium hydroxide, 2957923-80-5, 2957923-83-8; 18 h, reflux Experimental Protocols	Cyclometalated iridium NHC complexes containing self-isomerized ligands as catalysts for hydrosilylation and transfer hydrogenation reactions By: van Vuuren, Estefan; et al New Journal of Chemistry (2023), 47(11), 5203-5215.
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Scheme 13 (1 Reaction)

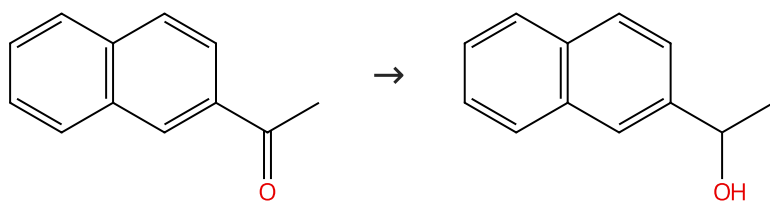
Steps: 1 Yield: 100%



31-513-CAS-22470180 Steps: 1 Yield: 100% 1.1 Reagents: Sodium formate Catalysts: Chloro[(1,2,3,4,5-η)-1,2,3,4,5-pentamethyl-2,4-cyclopentadien-1-yl]((<i>N</i>-phenyl-2-pyridinecarboxamidato-κ^N¹, κ^N²)iridium Solvents: Water; 15 h, 37 °C Experimental Protocols	Scope and limitations of reductive amination catalyzed by half-sandwich iridium complexes under mild reaction conditions By: Nguyen, Dat P.; et al Tetrahedron Letters (2020), 61(32), 152196.
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Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (92)

31-614-CAS-33034602

Steps: 1 Yield: 100%

Efficient transfer hydrogenation of ketones using molybdenum complexes by comprehensively verifying the auxiliary ligands

By: Wang, Zheng; et al

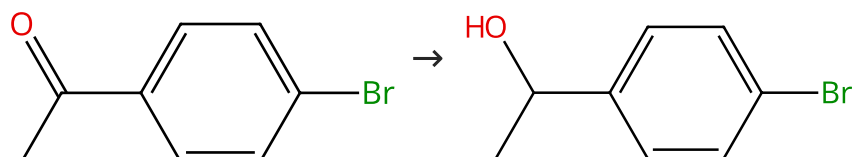
Dalton Transactions (2022), 51(29), 10983-10991.

1.1 Reagents: Sodium hydroxide
Catalysts: Sodium triethylborohydride, 2803321-31-3
Solvents: Isopropanol; 5 h, 90 °C

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

Suppliers (73)

31-513-CAS-23696964

Steps: 1 Yield: 100%

1,3,4-Oxadiazole-functionalized α-amino-phosphonates as ligands for the ruthenium-catalyzed reduction of ketones

By: Hkiri, Shaima; et al

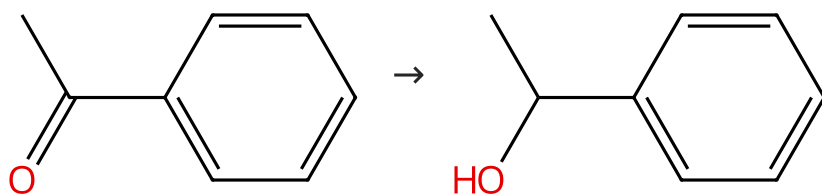
New Journal of Chemistry (2021), 45(25), 11327-11335.

1.1 Reagents: Sodium hydroxide
Catalysts: 2656431-71-7
Solvents: Isopropanol; 2 h, 100 °C

Experimental Protocols

Scheme 16 (5 Reactions)

Steps: 1 Yield: 100%



Suppliers (105)

Suppliers (71)

31-614-CAS-39254880

Steps: 1 Yield: 100%

Construction of Cu-Ru bimetallic catalyst for the selective catalytic transfer hydrogenation of carbonyl (C=O) in biomass-derived compounds

By: Cai, Bo; et al

Renewable Energy (2024), 222, 119833.

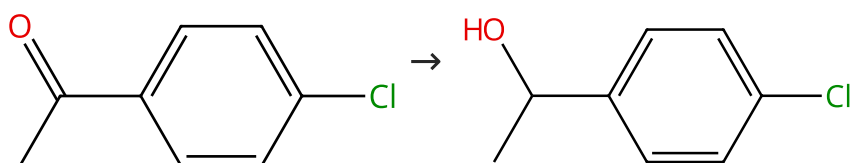
1.1 Reagents: Isopropanol
Catalysts: Zirconium dioxide, Sulfur trioxide, Silica, Copper alloy, nonbase, Cu,Ru; 12 h, 180 °C

Experimental Protocols

31-614-CAS-37500191 Steps: 1 Yield: 100% 1.1 Reagents: Isopropanol Catalysts: Potassium hydroxide, 2957923-80-5, 2957923-83-8; 18 h, reflux Experimental Protocols	Cyclometalated iridium NHC complexes containing self-isomerized ligands as catalysts for hydrosilylation and transfer hydrogenation reactions By: van Vuuren, Estefan; et al New Journal of Chemistry (2023), 47(11), 5203-5215.
31-614-CAS-32222218 Steps: 1 Yield: 100% 1.1 Reagents: Potassium hydroxide Catalysts: 2768853-49-0 Solvents: Isopropanol; rt → 80 °C; 3 h, 80 °C Experimental Protocols	Ru^{II} complexes of 1,2,3-triazole appended tertiary phosphines, [P(Ph){(o-C₆H₄)(1,2,3-N₃C(Ph)CH)}₂] and [P(Ph){o-C₆H₄(CC H)-(1,2,3-N₃-Ph)}₂]: highly active catalysts for transfer hydrogenation of carbonyl/nitro compounds and for α-alkylation of ketones By: Namdeo, Pavan K.; et al Dalton Transactions (2022), 51(17), 6795-6808.
31-513-CAS-23697055 Steps: 1 Yield: 100% 1.1 Reagents: Isopropanol, Sodium hydroxide Catalysts: 2656431-71-7; 2 h, 100 °C Experimental Protocols	1,3,4-Oxadiazole-functionalized α-amino-phosphonates as ligands for the ruthenium-catalyzed reduction of ketones By: Hkiri, Shaima; et al New Journal of Chemistry (2021), 45(25), 11327-11335.
31-513-CAS-20632255 Steps: 1 Yield: 100% 1.1 Reagents: Dodecyltrimethylammonium chloride, Sodium formate, Polyoxyethylene sorbitan monolaurate Catalysts: (SP-5-31)-Dichloro(3-phenyl-1<i>H</i>-inden-1-ylidene)bis (tricyclohexylphosphine)ruthenium Solvents: Acetophenone, Water; 85 °C; 12 h, 85 °C Experimental Protocols	Transfer-hydrogenation reactions of ketones/aldehydes in water using first generation ruthenium indenylidene olefin metathesis catalyst: One step towards sequential cross-metathesis/transfer hydrogenation reactions By: Ozturk, Bengi Ozgun; et al Molecular Catalysis (2020), 480, 110640pp..

Scheme 17 (2 Reactions)

Steps: 1 Yield: 100%



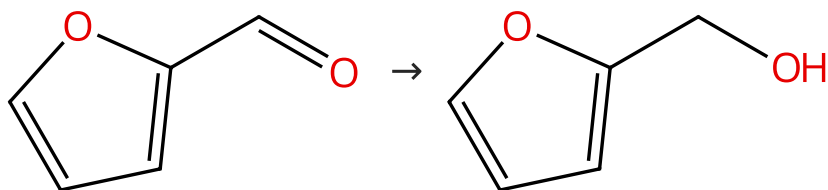
Suppliers (67)

Suppliers (67)

31-614-CAS-31770986 Steps: 1 Yield: 100% 1.1 Reagents: Sodium hydroxide Catalysts: 2763376-66-3 Solvents: Isopropanol; reflux Experimental Protocols	Synthesis and Reactivity of Ampy-Based Ruthenium(II) Catalysts for Transfer Hydrogenation of Ketones By: Amason, Edith K.; et al Organometallics (2022), 41(6), 686-697.
31-614-CAS-31770969 Steps: 1 Yield: 100% 1.1 Reagents: Sodium hydroxide Catalysts: 2763376-65-2 Solvents: Isopropanol; reflux Experimental Protocols	Synthesis and Reactivity of Ampy-Based Ruthenium(II) Catalysts for Transfer Hydrogenation of Ketones By: Amason, Edith K.; et al Organometallics (2022), 41(6), 686-697.

Scheme 18 (4 Reactions)

Steps: 1 Yield: 100%



Suppliers (81)

Suppliers (71)

31-614-CAS-43773059

Steps: 1 Yield: 100%

1.1 Reagents: Hydrogen
Catalysts: Graphene (oxide)
Solvents: Isopropanol; 4 h, 1 MPa, 150 °C

B, N Codoped Defective Reduced Graphene Oxide as a Highly Efficient Frustrated Lewis Pairs Catalyst for the Selective Hydrogenation of α,β -Unsaturated Aldehydes to Unsaturated Alcohols

By: Zhong, Linhao; et al

Journal of the American Chemical Society (2025), 147(4), 3840-3854.

31-614-CAS-35640445

Steps: 1 Yield: 100%

1.1 Reagents: Isopropanol
Catalysts: Aluminum, Ferrihydrite ($\text{Fe}_5(\text{OH})_9\text{O}_3$); 5 h, 1 MPa, 150 °C

Catalytic transfer hydrogenation of furfural to furfuryl alcohol over Al-containing ferrihydrite

By: Chen, Wenhai; et al

Journal of Industrial and Engineering Chemistry (Amsterdam, Netherlands) (2023), 119, 574-585.

Experimental Protocols

31-614-CAS-24870653

Steps: 1 Yield: 100%

1.1 Reagents: Isopropanol
Catalysts: Zirconium; 2 h, 120 °C

High-Efficiency Catalytic Transfer Hydrogenation of Biomass-Based 5-Hydroxymethylfurfural to 2,5-Bis(hydroxymethyl) furan over a Zirconium-Carbon Coordination Catalyst

By: He, Aiyong; et al

ACS Sustainable Chemistry & Engineering (2021), 9(46), 15557-15570.

Experimental Protocols

31-513-CAS-23352656

Steps: 1 Yield: 100%

1.1 Reagents: Formic acid
Solvents: 1,4-Dioxane; 1 h, 170 °C

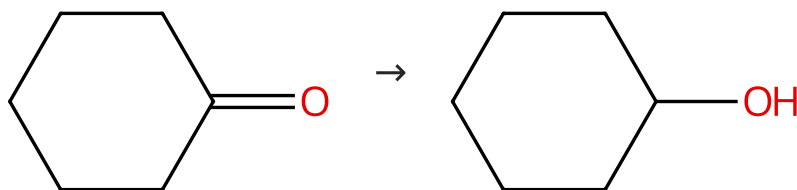
Catalytic transfer hydrogenation of biomass-derived furfural to furfuryl alcohol with formic acid as hydrogen donor over CuCs-MCM catalyst

By: Wang, Tao; et al

Chinese Chemical Letters (2021), 32(3), 1186-1190.

Scheme 19 (2 Reactions)

Steps: 1 Yield: 100%



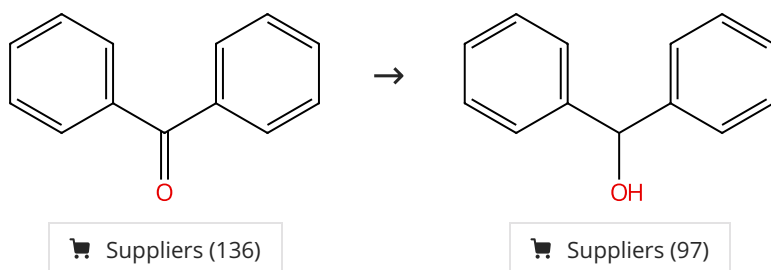
Suppliers (102)

Suppliers (81)

31-614-CAS-36607942 Steps: 1 Yield: 100% 1.1 Reagents: Potassium hydroxide Catalysts: 2918995-26-1 Solvents: Isopropanol; 2 h, reflux Experimental Protocols	Naphthyl-Derived Orthometalated Ru^{II}-NHC Complexes: Effect of the NHC Donors and/or Substitution Pattern on Their Synthesis and Catalytic Activity By: Bauri, Somnath; et al Organometallics (2020), 39(18), 3362-3374.
31-513-CAS-21446218 Steps: 1 Yield: 100% 1.1 Catalysts: Potassium <i>tert</i>-butoxide, 2409182-34-7 Solvents: Isopropanol, Water; 1 h, 80 °C Experimental Protocols	Strong solvent effects on catalytic transfer hydrogenation of ketones with [Ir(cod)(NHC)(PR₃)] catalysts in 2-propanol-water mixtures By: Orosz, Krisztina; et al Catalysts (2020), 10(1), 17pp..

Scheme 20 (1 Reaction)

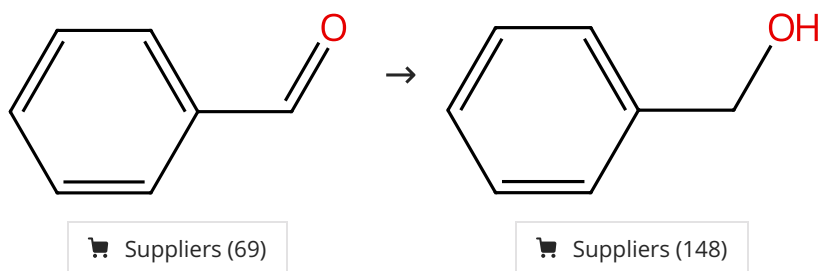
Steps: 1 Yield: 100%



31-513-CAS-20985681 Steps: 1 Yield: 100% 1.1 Reagents: Potassium hydroxide Catalysts: 2387645-44-3 Solvents: Isopropanol; 8 h, 80 °C Experimental Protocols	Ruthenium(II) complexes with pyridine-based Schiff base ligands: Synthesis, structural characterization and catalytic hydrogenation of ketones By: Buldurun, Kenan; et al Journal of Molecular Structure (2020), 1202, 127266.
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Scheme 21 (4 Reactions)

Steps: 1 Yield: 100%

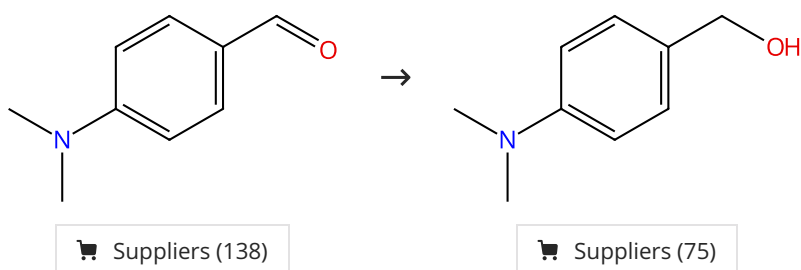


31-614-CAS-39254885 Steps: 1 Yield: 100% 1.1 Reagents: Isopropanol Catalysts: Zirconium dioxide, Sulfur trioxide, Silica, Copper alloy, nonbase, Cu,Ru; 12 h, 180 °C Experimental Protocols	Construction of Cu-Ru bimetallic catalyst for the selective catalytic transfer hydrogenation of carbonyl (C=O) in biomass-derived compounds By: Cai, Bo; et al Renewable Energy (2024), 222, 119833.
31-614-CAS-44207137 Steps: 1 Yield: 100% 1.1 Reagents: Isopropanol, Potassium hydroxide Catalysts: 3067927-59-4 Solvents: Water; 10 h, reflux 1.2 1 h, reflux Experimental Protocols	Iridium Complexes with Bidentate Pyridinylidene/N-Amidate Ligands for Transfer Hydrogenation Catalysis By: Finlay, Connal D.; et al Inorganic Chemistry (2024), 63(34), 15724-15734.

31-614-CAS-44207141 Steps: 1 Yield: 100% 1.1 Reagents: Isopropanol, Potassium hydroxide Catalysts: 3067927-61-8 Solvents: Water; 10 h, reflux 1.2 1 h, reflux Experimental Protocols	Iridium Complexes with Bidentate Pyridinylidene/N-Amidate Ligands for Transfer Hydrogenation Catalysis By: Finlay, Connal D.; et al Inorganic Chemistry (2024), 63(34), 15724-15734.
31-614-CAS-31110150 Steps: 1 Yield: 100% 1.1 Reagents: Potassium hydroxide Solvents: Isopropanol; 0.5 h, 80 °C	Visible Light Induced Reduction and Pinacol Coupling of Aldehydes and Ketones Catalyzed by Core/Shell Quantum Dots By: Xi, Zi-Wei; et al Journal of Organic Chemistry (2021), 86(3), 2474-2488.

Scheme 22 (2 Reactions)

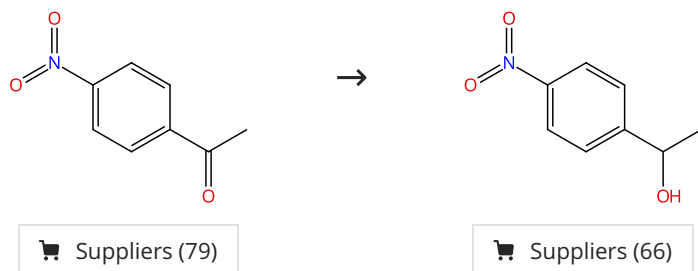
Steps: 1 Yield: 99%



31-614-CAS-41518337 Steps: 1 Yield: 99% 1.1 Reagents: Tris(4-methoxyphenyl)phosphine, 4-<i>tert</i>-Butylthiophenol, Water Catalysts: Iridium(1+), [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine-$\kappa N^1, \kappa N^1$]bis[3,5-difluoro-2-[5-(trifluoromethyl)-2-pyridinyl-κN]phenyl-κC]-, (OC-6-33)-, hexafluorophosphate(1-)(1:1) Solvents: Acetonitrile; 21 h, rt Experimental Protocols	Photoinduced catalytical reduction of carbonyl compounds using water as a hydrogen source By: Chen, Shenhao; et al Organic Chemistry Frontiers (2024), 11(19), 5415-5421.
31-614-CAS-31398123 Steps: 1 Yield: 99% 1.1 Reagents: 4-Methylbenzenethiol Catalysts: Cadmium sulfide, Cesium acetate Solvents: Toluene Experimental Protocols	Visible Light Induced Reduction and Pinacol Coupling of Aldehydes and Ketones Catalyzed by Core/Shell Quantum Dots By: Xi, Zi-Wei; et al Journal of Organic Chemistry (2021), 86(3), 2474-2488.

Scheme 23 (2 Reactions)

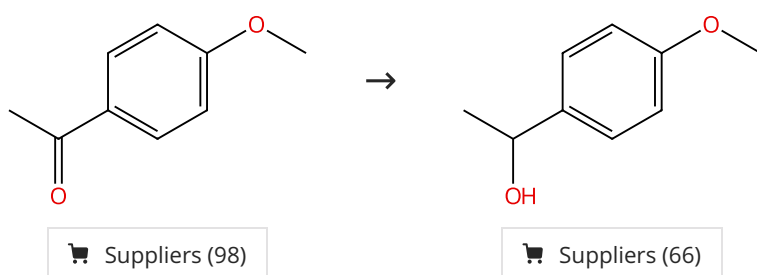
Steps: 1 Yield: 99%



31-614-CAS-42483378	Steps: 1 Yield: 99%	A cobalt-based metal-organic framework as a sustainable catalyst for base-free transfer hydrogenation of biomass-derived carbonyl compounds
1.1 Catalysts: 3068691-39-1 Solvents: Isopropanol; 2.5 h, 60 °C		By: Aashish; et al
Experimental Protocols		RSC Sustainability (2024), 2(11), 3409-3423.
31-614-CAS-33034572	Steps: 1 Yield: 99%	Efficient transfer hydrogenation of ketones using molybdenum complexes by comprehensively verifying the auxiliary ligands
1.1 Reagents: Sodium hydroxide Catalysts: Sodium triethylborohydride, 2803321-31-3 Solvents: Isopropanol; 2 h, 90 °C		By: Wang, Zheng; et al
Experimental Protocols		Dalton Transactions (2022), 51(29), 10983-10991.

Scheme 24 (3 Reactions)

Steps: 1 Yield: 99%



31-614-CAS-37701135	Steps: 1 Yield: 99%	Triazole backbone ligand in an unprecedented efficient manganese catalyst for use in transfer hydrogenation
1.1 Reagents: Isopropanol, Potassium <i>tert</i> -butoxide Catalysts: Bromopentacarbonylmanganese, 2,2'-(1-Phenyl-1 <i>H</i> -1,2,4-triazole-3,5-diyl)bis[pyridine]; 2 h, 80 °C		By: Liang, Qianqian; et al
Experimental Protocols		Science China: Chemistry (2023), 66(7), 2028-2036.
31-614-CAS-37701132	Steps: 1 Yield: 99%	Triazole backbone ligand in an unprecedented efficient manganese catalyst for use in transfer hydrogenation
1.1 Reagents: Isopropanol, Potassium <i>tert</i> -butoxide Catalysts: Bromopentacarbonylmanganese, 2982788-05-4; 2 h, 80 °C		By: Liang, Qianqian; et al
Experimental Protocols		Science China: Chemistry (2023), 66(7), 2028-2036.
31-513-CAS-22733873	Steps: 1 Yield: 99%	Synthesis and catalytic applications of Ru and Ir complexes containing N,O-chelating ligand
1.1 Reagents: Potassium hydroxide Catalysts: Chloro[(1,2,3,4,5-η)-1,2,3,4,5-pentamethyl-2,4-cyclopentadien-1-yl](2-pyridinecarboxylato-κ <i>N</i> ¹ ,κ <i>O</i> ²)iridium Solvents: Isopropanol; 0.5 h, 82 °C		By: Pakyapan, Bilge; et al
Experimental Protocols		Journal of Organometallic Chemistry (2020), 925, 121486.